USED PAPERSAUG/SEP-2021

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Code No: 156DY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2021
QUANTITATIVE ANALYSIS FOR BUSINESS DECISIONS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the graphical method of solving a linear programming problem.
b) Solve by using graphical method
Max $Z = 4x_1 + 3x_2$
Subject to
$4x_1 + 3x_2 \leq 24$, $x_1 \leq 4.5$, $x_2 \leq 6$ and
$x_1 \geq 0$, $x_2 \geq 0$. [7+8]

2. Solve the following linear programming problem using Simplex method
Maximize $Z = 5x_1 + 7x_2$
Subject to
$x_1 + x_2 \leq 4$
$3x_1 - 8x_2 \leq 24$
$10x_1 + 7x_2 \leq 35$
And $x_1 \geq 0$, $x_2 \geq 0$. [15]

3. Using North-West Corner rule for initial basic feasible solution, obtain an optimum basic feasible solution to the following problem. [15]

	To			Available
	7	3	4	2
From	2	1	3	3
	3	4	6	5
Demand	4	1	5	10

4.a) A car hire company has one car at each of five depots a, b, c, d and e. a customer requires a car in each town namely A, B, C, D and E. Distance (kms) between depots (origins) and towns (destinations) are given in the following distance matrix

	a	b	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185
D	50	50	80	80	110
E	55	35	70	80	105

b) Explain the concept of Travelling salesman problem. [8+7]

5.a) What are situations that make replacement of capital cost items?
b) Explain, with examples the failure mechanism of items. [8+7]

6. The cost of a machine is Rs. 6,100 and its scrap value is Rs. 10. The maintenance costs found from experience are as follows:

Year	: 1	2	3	4	5	6	7	8
Maintenance cost(Rs)	: 100	250	400	600	900	1200	1600	2000

When should the machine replaced? [15]

7.a) Define i) Competitive game
ii) Payoff matrix
iii) Saddle pint
iv) Two-person zero sum game
b) Solve the following game and find the optimal strategies: [7+8]

	B1	B2	B3	B4
A1	3	-5	0	6
A2	-4	-2	1	2
A3	5	4	2	3

8.a) Explain in brief, the main characteristics of the ‘ Queueing system’.
b) A Television repairman finds that the time spent on his jobs has an exponential distribution with mean 30 minutes. If he repairs the sets in the order in which they come in, and if the arrivals of sets are approximately Poisson with an average rate of 10 per 8 hours day which is the repairs man idle time each day? Find the expected number of units in the system and in the queue. [8+7]

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